PRODUCT REQUIREMENTS DOCUMENT

Fleet Maintenance & Telematics Diagnostic Agent

PRD v1.0 | Prepared for project review

Product concept

An AI-assisted fleet maintenance platform that uses vehicle diagnostic trouble codes (DTCs), odometer readings and maintenance history to help fleet teams identify issues, prioritize maintenance and initiate service or spare-parts workflows.

Proposed technology stack

Frontend: Next.js | Backend: Node.js + Express.js | Database: MongoDB

Document status: Draft for project-head review and scope confirmation. Requirements and integrations are proposed, not yet approved.

Prepared as an initial product definition for MVP planning.

1. Document Overview

This document translates the supplied project idea into a reviewable product scope. It defines the problem, users, MVP capabilities, workflows, technical direction, assumptions and open decisions. It is intended to obtain alignment before implementation begins.

Field	Details
Product	Fleet Maintenance & Telematics Diagnostic Agent
Version	1.0 – Draft
Primary audience	Project Head, Product/Engineering Team
Product type	B2B fleet-management SaaS
Initial platform	Responsive web application
Proposed stack	Next.js, Node.js, Express.js, MongoDB
MVP data approach	Manual entry and simulated vehicle diagnostic data until integrations are confirmed

1.1 Project idea supplied

“Reads engine diagnostic codes (OBD-II), odometer readings, and historical failure patterns to automatically book maintenance slots and order spare parts.” The concept also proposes a subscription price of **$15 per truck per month** (illustratively $1,500/month for a 100-vehicle fleet). Pricing is treated as a concept input, not a validated commercial decision.

2. Executive Summary

Fleet operators need to keep commercial vehicles available while controlling repair costs and unplanned downtime. Diagnostic trouble codes, mileage and service records can provide useful maintenance signals, but teams may have to interpret those signals, decide what to do next and coordinate workshops and parts separately.

The proposed product brings these steps into one workflow. It will maintain vehicle records, collect or accept diagnostic information, use an AI-assisted agent to explain possible issues and recommend next actions, and create maintenance or parts requests. For the MVP, external actions will be represented as requests requiring user approval; real booking and purchasing will only be claimed as completed after an external provider confirms them.

The product should be an operational assistant—not simply a dashboard or a chatbot. Its value is the connected workflow from vehicle signal to reviewed action.

3. Problem Statement & Opportunity

Fleet managers and maintenance teams may face fragmented vehicle data, manual fault-code interpretation, inconsistent service records and time-consuming coordination with service providers and parts suppliers. Delayed action can increase the risk of vehicle downtime and disrupt fleet operations.

The product opportunity is to make maintenance information easier to interpret and convert it into traceable, approval-based workflows. It should help teams answer: What is happening? How urgent is it? What evidence supports the recommendation? What action should happen next? Has that action actually been confirmed?

4. Product Vision, Goals & Non-Goals

4.1 Vision

Provide fleet teams with a clear, explainable and action-oriented maintenance assistant that connects vehicle health signals with maintenance planning.

4.2 MVP goals

- Provide secure access to an organization’s fleet workspace.
- Create and maintain vehicle profiles and odometer readings.
- Record diagnostic trouble codes and maintenance history.
- Generate structured AI-assisted explanations and recommendations from available data.
- Prioritize issues using transparent rules and available history.
- Create service requests and spare-parts requests with approval and status tracking.
- Keep an auditable record of analyses, recommendations and user actions.

4.3 Non-goals for the initial MVP

- Direct integration with every OBD-II device or telematics provider.
- Guaranteed prediction of component failure or exact remaining useful life.
- Unattended purchasing or payment processing.
- Guaranteed automatic workshop booking without a confirmed provider integration.
- A full ERP, accounting, GPS-routing or fleet-dispatch platform.

5. Target Users & Roles

User / Role	Primary needs	MVP permissions
Fleet Manager	Fleet overview, issue prioritization, approvals, service status and reports.	Organization-wide view; manage vehicles; approve/reject requests.
Maintenance Staff	Review diagnostics, update service records and prepare maintenance requests.	View assigned fleet data; add records; create requests.
Mechanic / Service Provider (future or limited MVP)	Receive vehicle issue details and update service progress.	Provider-side access only if approved and implemented.
Organization Admin	Manage users, roles and organization settings.	Invite/manage users; configure organization.

6. Core User Journeys

A. Register and set up fleet

An authorized user registers or signs in, creates or joins an organization, and adds vehicle profiles.

B. Record a vehicle issue

A user selects a vehicle and enters a DTC, odometer reading, date/time and optional notes or source details.

C. Run diagnostic analysis

The user requests analysis. The backend gathers the relevant DTC description and vehicle history, applies deterministic checks and asks the AI service for a structured explanation.

D. Review recommendation

The user sees possible causes, severity, evidence, uncertainty, recommended checks and suggested next action.

E. Request service or parts

The user reviews and approves a draft service request or parts request. The system tracks it as pending until a provider or authorized user confirms the outcome.

F. Close the maintenance loop

After work is completed, a user records the service outcome, parts used, costs if available and next service information.

7. Functional Requirements

Priority definitions: **P0** = required for MVP; **P1** = valuable if time permits; **Future** = post-MVP.

ID	Requirement	Priority	Acceptance criteria
FR-01	Authentication & authorization	P0	Users can register/sign in/sign out. Passwords are securely hashed. Protected APIs enforce authentication and role/organization access.
FR-02	Organization & user management	P0	A user's data is scoped to the correct organization. Admin can manage basic users/roles, subject to MVP scope.
FR-03	Fleet dashboard	P0	Dashboard displays fleet totals and counts of open issues, maintenance due and pending requests using persisted data.
FR-04	Vehicle management	P0	Authorized users can create, view, update and archive vehicle profiles. Required fields are validated.
FR-05	Diagnostic record capture	P0	Users can add a DTC, odometer value, timestamp, source and notes. Records are linked to a vehicle and retained in history.
FR-06	Diagnostic history	P0	Users can view a vehicle's diagnostic records in chronological order and open a record's details.
FR-07	AI diagnostic analysis	P0	The agent returns structured possible causes, explanation, severity, evidence, uncertainty and recommended checks. It must not present an unverified cause as fact.
FR-08	Maintenance risk assessment	P0	The system assigns a rule-based priority using documented conditions and available history. It explains the factors and labels estimates clearly.
FR-09	Maintenance history	P0	Users can add and view completed maintenance events, date, odometer, work performed and optional cost/parts.
FR-10	Maintenance recommendation	P0	The system can create a suggested next action with reason, priority and review status.
FR-11	Service request workflow	P0	Users can create, review, approve/reject and track a service request. Status changes are recorded. No booking is marked confirmed without confirmation.
FR-12	Parts recommendation/request	P0	The system can show suggested parts or a parts request with reason and quantity where supported. A user must approve before an order is placed.
FR-13	Notifications	P1	In-app notifications are created for important issue and request status changes.
FR-14	Reports & analytics	P1	Users can view basic summaries such as issue counts, maintenance history and request status.
FR-15	External telematics integration	Future	Connect supported OBD-II/telematics providers after API, hardware, access and data-format decisions are confirmed.

8. AI Agent Requirements & Guardrails

The AI agent should orchestrate a small set of well-defined backend tools. It should not have unrestricted access to the database or external services.

Agent capability	Expected behavior
Diagnostic explanation	Explain the supplied DTC in plain language; show possible causes and checks, not a definitive diagnosis.
History context	Use the selected vehicle's available diagnostic and maintenance history; identify missing data.
Priority assessment	Combine documented deterministic rules with available context. Show why a priority was assigned.

Agent capability	Expected behavior
Maintenance planning	Draft a recommended action and service request for user review.
Parts assistance	Suggest a part category or catalog match only when supported by reliable catalog data; avoid unsupported part numbers.
Tool execution	Validate inputs and user permissions server-side. Require approval for external booking or purchasing.
Auditability	Store input references, output, timestamp, model/service metadata where available, and approval/action history.

8.1 Safety and reliability rules

- A diagnostic trouble code is an indicator, not proof that a specific component has failed. For example, P0301 can indicate a cylinder 1 misfire; inspection is needed to establish the cause.
- The agent must distinguish observed facts, likely explanations and uncertain hypotheses.
- If data is missing, stale or contradictory, the agent must say so and avoid overconfident recommendations.
- Critical safety-related warnings should advise prompt professional inspection and must not encourage continued operation where unsafe.
- Only backend-validated, authorized tool calls may create or update records.
- External booking/order status must reflect provider confirmation, not merely an AI response.

9. MVP Scope & Delivery Boundary

The recommended MVP demonstrates the complete workflow using sample or manually entered data. This allows the team to validate the product experience before committing to hardware and vendor integrations.

Included in MVP	Deferred / dependent on confirmation
Authentication and organization-scoped access	Live integrations with OBD-II hardware and telematics vendors
Vehicle CRUD and fleet dashboard	Continuous real-time vehicle streaming
Manual/simulated DTC and odometer input	GPS tracking, route optimization and dispatch
Diagnostic history and maintenance records	Trained predictive ML models using labeled failure datasets
AI-assisted explanation and rule-based prioritization	Guaranteed failure prediction / remaining-life estimates
Approval-based service and parts request workflows	Live provider booking and supplier checkout/payment
Basic request tracking and audit history	Advanced BI, invoicing, warranty and ERP integrations

MVP completion should be judged against an agreed feature checklist and demonstration scenario. A percentage such as “45% complete” should be defined by the project head (for example, feature-weighted scope, milestone completion or effort estimate) before it is used as an acceptance measure.

10. Data Inputs & Initial Data Strategy

Data	MVP source	Notes
Vehicle profile	Manual entry	Registration/fleet identifier, make, model, year and other agreed fields.
Diagnostic trouble codes	Manual entry / seeded sample data	Store code, description if known, timestamp and source.
Odometer	Manual entry / simulated feed	Store reading, unit and timestamp; validate that readings are plausible.
Maintenance history	Manual entry / sample records	Service date, odometer, work performed, parts and optional cost.

Data	MVP source	Notes
Parts catalog	Seeded sample catalog	Clearly label sample inventory; do not imply supplier availability.
Service providers	Seeded sample providers	Requests remain simulated/pending unless an actual integration is approved.

Future integration readiness: define adapter/service boundaries so that telematics providers, OBD-II devices, service networks and parts suppliers can be added without redesigning core vehicle and maintenance entities.

11. Proposed Information Architecture & Screens

Screen / Module	Purpose
Login / Register	Secure user access.
Organization setup	Create or join a fleet workspace; basic user/role setup.
Dashboard	Fleet overview, issue priorities and pending actions.
Vehicles	Search, filter and manage vehicle records.
Vehicle detail	Vehicle profile, odometer, diagnostics and maintenance timeline.
Diagnostics	Add and inspect diagnostic records.
AI Diagnostic Agent	Run analysis and review explanation, evidence and recommendations.
Maintenance	View maintenance history and suggested next actions.
Service Requests	Create, approve and track workshop/service requests.
Parts Requests	Review suggested parts and track approval/request status.
Notifications / Settings	In-app alerts and basic account/organization settings.

12. Proposed Technical Architecture

This is a proposed implementation direction based on the requested stack. Final module boundaries and deployment choices should be confirmed by the engineering lead.

Layer	Proposed responsibility
Frontend – Next.js (JavaScript)	Responsive UI, route protection, forms, dashboard, reusable components and API client.
Backend – Node.js + Express.js	REST API, authentication, authorization, validation, business rules, AI orchestration and tool execution.
Database – MongoDB + Mongoose	Persist organizations, users, vehicles, diagnostics, maintenance events, analyses and requests.
AI service	LLM integration for structured explanations and tool selection; backend retains control of permissions and execution.
External adapters (future)	Telematics, service provider and parts supplier integrations behind dedicated interfaces.

12.1 Suggested REST resource groups

`/api/auth` · `/api/organizations` · `/api/users` · `/api/vehicles` · `/api/diagnostics` · `/api/maintenance` · `/api/agent` · `/api/service-requests` · `/api/parts` · `/api/notifications`

12.2 Initial MongoDB collections (proposed)

Users, Organizations, Vehicles, DiagnosticRecords, MaintenanceHistory, AIAnalyses, MaintenanceRequests, ServiceProviders, Parts, PartsOrders, Notifications and AuditLogs. Collection design should enforce organization ownership and appropriate indexes.

13. Non-Functional Requirements

Area	Requirement
Security	Hash passwords; validate and sanitize inputs; enforce authentication, role checks and organization-level data isolation; protect secrets in environment variables; use HTTPS in deployment.
Privacy	Collect only required vehicle and user data. Define retention and access rules before production use.
Performance	Paginate large lists; index frequently queried fields; avoid loading full fleet histories on the dashboard; set API timeouts.
Reliability	Use consistent error responses, server-side validation, logging and safe failure behavior for AI/provider outages.
Usability	Responsive layouts, clear loading/empty/error states, accessible labels, keyboard navigation and readable status indicators.
Maintainability	Use reusable components, modular backend services, clear naming, documented API contracts and environment configuration.
Testing	Test authentication, authorization, validation, core CRUD, agent output schema, approval workflows and key user journeys.

14. Security, Permissions & Audit

- Every vehicle, diagnostic, maintenance and request record must be associated with an organization.
- The backend must verify that the signed-in user is allowed to access the requested organization and resource.
- Validate request bodies, query parameters and path identifiers; never trust client-side role checks alone.
- Protect AI tool execution with allowlisted tools, schema validation, permission checks and approval gates.
- Record who created, approved, rejected or changed a service/parts request, and when.
- Do not expose API keys, database credentials or internal stack traces to the browser.

15. Success Metrics (MVP)

The following are proposed measures. Baselines and targets should be agreed after the first usable build.

Metric	How it can be measured
Fleet setup success	Percentage of test organizations able to add a vehicle and view it on the dashboard.
Diagnostic workflow completion	Percentage of seeded/manual DTC records that can be analyzed and reviewed end-to-end.
Recommendation explainability	Reviewer checks whether each recommendation shows evidence, uncertainty and next steps.
Request workflow completion	Percentage of service/parts requests that move through the expected approval/status lifecycle.
Data integrity	Number of invalid or cross-organization record accesses accepted (target: zero).
Usability	Completion of key tasks by a small set of representative users without assistance.

16. Risks & Mitigations

Risk	Mitigation
No confirmed OBD-II/telematics access	Start with manual/simulated data; keep integration adapters separate.
AI gives an incorrect diagnosis	Use cautious language, structured outputs, rule checks, evidence display and mechanic review.

Risk	Mitigation
No live booking or supplier API	Implement approval-based request workflows and label them as requests, not confirmed bookings/orders.
Insufficient historical failure data	Do not claim predictive accuracy; use transparent rule-based prioritization in MVP.
Scope becomes too large	Protect the MVP boundary and prioritize one complete end-to-end workflow.
Sensitive fleet data exposure	Apply organization isolation, least privilege, secure secrets and audit logging.

17. Competitor & Market Context

Public product information indicates that this is an established fleet-maintenance software category. The following products provide relevant reference points; this is a directional feature review, not an exhaustive market study.

Product	Publicly described capability relevant to this concept
Geotab – Predictive Maintenance / maintenance tools	Telematics-based maintenance insights, diagnostic faults, vehicle health signals and maintenance workflows.
Motive – Maintenance	Fault-code monitoring and explanations, maintenance scheduling/work orders, service history and parts/inventory capabilities across its product offering.
Fleetio – AI Service Advisor	AI-assisted maintenance decisions, fault monitoring, issue prioritization and work-order-oriented workflows.
Samsara – maintenance solutions	Fleet maintenance and diagnostics capabilities within a broader connected-operations platform.

Competitive implication: a dashboard, DTC lookup or generic AI chat alone is unlikely to be a strong differentiator. The proposed product should focus on a coherent, explainable workflow from signal → context → recommendation → human approval → tracked action. A simpler experience for small and mid-sized fleets is a hypothesis to validate, not an established market gap.

Reference starting points: Geotab (geotab.com), Motive (gomotive.com), Fleetio (fleetio.com) and Samsara (samsara.com). Product capabilities and packaging may vary by region, plan and hardware.

18. Commercial Concept

The supplied concept proposes a subscription of **$15 per truck per month**, with an illustrative 100-truck fleet producing $1,500 in monthly recurring revenue. This arithmetic is correct, but the price itself has not been validated against customer willingness to pay, integration costs, support costs, competitor packaging or regional purchasing power.

Pricing should be treated as a hypothesis for discovery. Potential packaging variables include number of active vehicles, diagnostic/AI usage, integrations, support level and advanced reporting.

19. Proposed MVP Acceptance Demonstration

A successful MVP demo should complete this scenario using a sample fleet and clearly labeled simulated data:

- A fleet manager signs in and sees the organization dashboard.
- The manager opens a vehicle profile and reviews its odometer and maintenance history.
- The manager adds a diagnostic code and odometer reading.
- The agent returns a structured explanation, possible causes, severity, evidence, uncertainty and recommended checks.
- The manager reviews the recommendation and creates a service request.
- The manager approves the request; the system records it as approved/pending provider confirmation—not as a confirmed booking.

• A user records a completed maintenance event and the vehicle timeline updates.

20. Assumptions & Decisions Required from Project Head

The following items are unresolved and should be confirmed before implementation scope is finalized.

Open decision	Current working assumption / question
Vehicle data source	Unknown. MVP assumes manual entry and simulated data. Confirm whether hardware or telematics API access will be provided.
Target geography	Unknown. Keep currency, units, locale and provider configuration flexible; confirm initial customer geography.
External service booking	Unknown. Confirm whether a provider API or partner is available; otherwise MVP creates service requests only.
Spare-parts ordering	Unknown. Confirm catalog/supplier access; otherwise use a sample catalog and approval-based request flow.
AI provider/model	Not finalized. Confirm approved provider, API access, budget, data policy and expected structured output.
User roles	Confirm whether MVP requires only fleet manager/admin or also maintenance staff and provider accounts.
45% delivery milestone	Confirm how 45% completion is measured: feature-weighted scope, milestone checklist, effort or another definition.
Deployment & environments	Confirm hosting, database environment, repository access, secrets management and review process.

21. Suggested MVP Work Breakdown (Provisional)

This is a suggested planning allocation only. It is not a company-approved definition of the 45% milestone.

Workstream	Indicative share of initial milestone
Project setup, architecture and API conventions	5%
Authentication and organization access	5%
Navigation and dashboard shell	10%
Vehicle management and diagnostic record capture	10%
Initial AI diagnostic workflow	10%
Maintenance recommendations, request flow and testing	5%
Total	45%

22. Recommendation & Next Steps

Approve or revise the MVP boundary and resolve the key integration assumptions before development starts. Once approved, the team can convert this PRD into user stories, API contracts, database schemas, UI wireframes and a prioritized implementation backlog.

- Project head reviews the goals, MVP scope and acceptance demonstration.
- Confirm the 45% milestone definition and expected delivery date.
- Confirm data source, geography, AI provider and external integration availability.
- Approve user roles, security expectations and deployment environment.

- Break approved requirements into development tasks and begin implementation.

End of document · PRD v1.0 · Draft for review